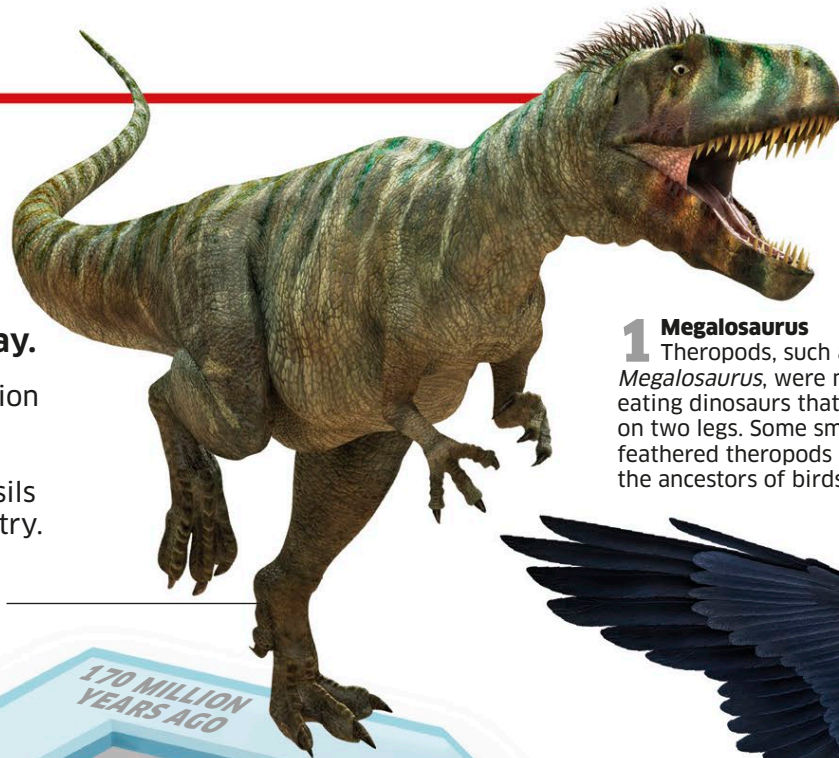


The fossil record

Fossils from prehistoric times show just how much life has changed across the ages, and how ancient creatures are related to the organisms on Earth today.

Life has been evolving on our planet for more than four billion years—ever since it was just a world of simple microbes. Across this vast expanse of time, more complex animals and plants developed. Traces of their remains—found as fossils in prehistoric rocks—have helped us to work out their ancestry.



1 Megalosaurus Theropods, such as *Megalosaurus*, were meat-eating dinosaurs that walked on two legs. Some smaller, feathered theropods were the ancestors of birds.

Like most birds, theropods had feet with three forward-pointing toes, and hollow bones.



The origins of birds

Fossilized skeletons show us that the first prehistoric birds were remarkably similar to a group of upright-walking dinosaurs. From these fossils, it is possible to see how their forelimbs evolved into wings for flight, and how they developed the other characteristics of modern birds.

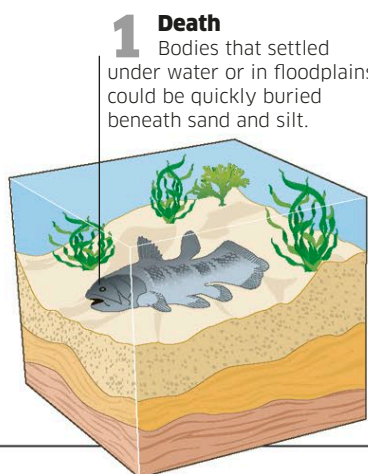


Archaeopteryx fossil

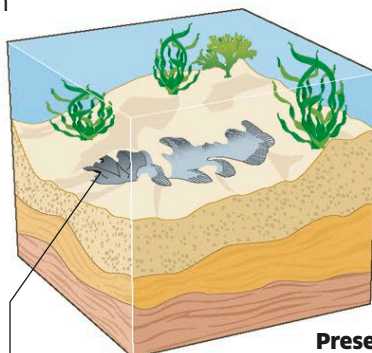
This fossil of *Archaeopteryx* has been preserved in soft limestone. Around the animal's wing bones, the imprints left by the feathers are clearly visible.

How fossils form

Fossils are the remains or impressions of organisms that died more than 10,000 years ago. Some fossils have recorded what is left of entire bodies, but usually only fragments, such as parts of a bony skeleton, have survived.

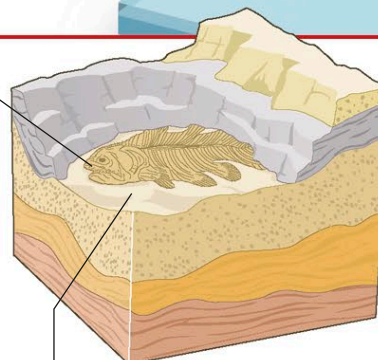


1 Death Bodies that settled under water or in floodplains could be quickly buried beneath sand and silt.



2 Burial Layers of sediment cover the body and build up into rock on top of it.

Skeletons and other hard parts are more likely to leave an impression than soft tissues.



3 Reveal Millions of years later, movements of Earth's crust cause rocks to move upward, exposing the fossil on dry land.

Preserved in time

When organisms in the prehistoric world died, their bodies were more likely to be preserved if they were quickly buried. Rotting under layers of sediment, the body slowly turned into mineral, until the resulting fossil was exposed by erosion.

Over millions of years, groups of organisms split up as they evolve and become adapted to new environments or situations.